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## **SIMULATION OF GAS TURBINE ENGINES**

Belo Horizonte  
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## **Simulation of Gas Turbine Engines**

A thesis submitted to the Faculty of the School of Engineering of the Universidade Federal de Minas Gerais in partial fulfillment of the requirements for the degree of Bachelor of Science in Aerospace Engineering.

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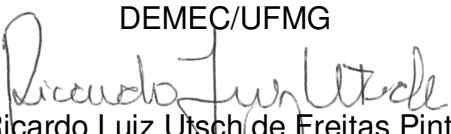
## **"Simulation of Gas Turbine Engines"**

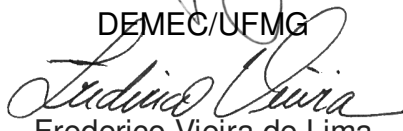
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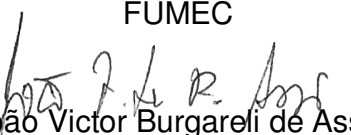
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*To my family*

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DON'T PANIC.

*-The Hitchhiker's Guide to the Galaxy*

# Abstract

The simulation of gas turbine engines is of vital importance to their design process and to the design of their control systems. In order to capture behaviour not shown by a simple thermodynamic cycle model, such as transients, unchoked turbine inlet, and instabilities, as well as to increase the model fidelity, it is necessary to develop a component based engine model. This work develops such engine model for the simple case of a single spool turbojet, with a single stage centrifugal compressor and single stage axial turbine. This configuration is typical of very small gas turbines. No experimental data is need as input to the model. It is based on first principles and semi-empirical corrections available in the literature. This work contains nonlinear models for a centrifugal compressor, axial turbine, nozzle, and the engine operating statically and dynamically. Surge is considered. The models are applied for the Jet Munts VT-80 engine and the results are compared with experimental data.

## Resumo

A simulação de turbinas a gás é essencial para seu projeto e para o projeto dos seus sistemas de controle. Para simular características deste tipo de motor que não são capturadas por um simples modelo de ciclo termodinâmico, e também para aumentar a fidelidade da simulação, é necessário desenvolver um modelo de componentes. Este trabalho desenvolve um modelo desse tipo para o caso de um motor turbojato de um único eixo, com um compressor centrífugo de um estágio e uma turbina axial também de um único estágio. Essa configuração é típica para mini-turbinas a gás. Nenhum dado experimental é necessário como entrada para o modelo, pois ele é baseado em leis de conservação e correções semi-empíricas encontradas na literatura. Este trabalho contém modelos não lineares para compressores centrífugos, turbinas axiais, bocais, e motores completos operando em regime permanente ou transiente. A purga do compressor é levada em conta. Esses modelos são aplicados para o caso do motor Jet Muntz VT-80 e os resultados comparados com dados experimentais.



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# Acronyms

**DAE** differential-algebraic equation.

**DOF** degrees of freedom.

**ECU** Engine Control Unit.

**EGT** Exhaust Gas Temperature.

**FADEC** Full Authority Digital Engine Controller.

**GSP** Gas Turbine Simulation Program.

**GTE** gas turbine engine.

**IGV** inlet guide vane.

**ODE** ordinary differential equation.

**PDE** partial differential equation.

**T-MATS** Toolbox for the Modeling and Analysis of Thermodynamic Systems.

# List of Symbols

$A$  area.  
 $B$  surge parameter.  
 $C'$  compressor characteristic derivative.  
 $F$  thrust.  
 $I$  moment of inertial of the spool.  
 $L$  length.  
 $M_b$  blade stagnation Mach number.  
 $M$  Mach number.  
 $P$  pressure.  
 $R$  gas constant.  
 $T'$  turbine characteristic derivative.  
 $T$  temperature.  
 $U$  rotor speed.  
 $V$  velocity in the inertial frame of reference.  
 $W$  velocity in the rotating frame of reference.  
 $Y$  pressure loss coefficient.  
 $Z$  number of blades.  
 $\Delta$  difference.  
 $\alpha$  angle in the inertial frame of reference.  
 $\beta$  angle in the rotating frame of reference.  
 $\delta$  angular deviation.  
 $\dot{m}$  mass flow rate.  
 $\eta_f$  combustion efficiency.  
 $\eta_p$  polytropic efficiency.  
 $\eta_t$  isentropic efficiency.  
 $\gamma$  specific heat ratio.  
 $\mathbb{R}$  set of real numbers.  
 $\nabla$  jacobian.  
 $\omega$  angular speed.  
 $\phi$  flow coefficient.  
 $\psi$  load coefficient.  
 $\rho$  density.  
 $\sigma$  slip factor.  
 $\tau$  torque.  
 $\xi$  blade geometric incidence.  
 $a$  sonic speed.

$c_p$  specific heat at constant pressure.  
 $c_v$  specific heat at constant volume.  
 $c$  chord.  
 $h$  specific enthalpy.  
 $r$  rotor radius.  
 $s$  solidity.  
 $t$  thickness.  
 $\forall$  burner volume.  
**FHV** fuel lower calorific value.  
**MFP** mass flow parameter.

## Subscripts

0 stagnation.  
 1 engine inlet station.  
 2 compressor inlet station.  
 3 compressor exit station.  
 4 turbine inlet station.  
 5 turbine exit station.  
 9 nozzle exit station.  
 $a$  ambient.  
 $c$  compressor.  
 $e$  exit.  
 $f$  fuel.  
 $i$  inlet.  
 $p$  burner (from plenum or pipe).  
 $r$  radial direction.  
 $t$  turbine.  
 $u$  direction of rotor velocity (i.e. tangential).  
 $x$  meridional direction.  
**actual** measured in practice.  
**euler** calculated by Euler's equation.  
**isen** isentropic.  
**opt** optimal.  
**Superscripts**  
 \* sonic.  
 $\cdot$  time derivative.

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# 1 Introduction

*This was the very reason why you were brought to Narnia, that by knowing me here for a little, you may know me better there.*

*-Aslan*

The development of gas turbine engines (GTEs) is historically related to the development of their control systems, nowadays represented by a Full Authority Digital Engine Controller (FADEC) or an Engine Control Unit (ECU). GTEs often operate their best when close to their thermal, mechanical or stability limits, which makes a control system essential. This control system is responsible for translating the pilot requests in terms of required thrust into inputs to the underlying turbomachinery in the form of changes in fuel flow or variable geometry. The design goal for an engine control system is to provide the fastest response possible to changes in pilot input while keeping the engine operating stably and within the designed temperature, pressure and speed limits (AUSTIN SPANG III; BROWN, 1999).

To design such a control system, it is desirable to have a computer model of the engine capable of reproducing all relevant phenomena of dynamical behavior. There are two basic types of GTE models. The most elementary one is the thermodynamic cycle model, which does not consider the mechanical aspects of the turbomachinery, taking them as idealized thermodynamical processes. It relies heavily on the assumption that the turbine inlet is choked, which is not always true, especially for operating conditions far from the design point and for poorly designed engines. The next level is the component model, which relies on theoretical or empirical models for the compressors and turbines in an engine (the so called maps) and applies conservation laws among them to calculate the engine performance. The goal of the present work is to develop a component based engine model and the model of the components themselves.

The development of a component level engine model is a three stage process. Firstly, individual component models must be developed, then they must be matched so that the steady state operation can be calculated. Finally, the dynamical behavior of the engine can be taken into account, through the introduction of a dynamical system comprising three states: spool speed, burner density and mass flow rate.

This work aims to go through all these steps without requiring experimental data from the particular design being analysed. This is necessary during the engine design, or when the time and costs involved with testing can not be afforded. Even when experimental data is available, it may not be enough to fully characterize the components, since it may be hard to operate the components in all points of their maps while in a test bed, and experimental measurements are inherently noisy. In this case, it is best to use the experimental data to calibrate and validate the theoretical model. The models employed here are based on first principles whenever possible, with the inclusion of semi-empirical corrections when necessary.

This work favors breadth instead of depth: for example, the models are presented for a single spool turbojet with a single stage centrifugal compressor and also single stage axial turbine; and compressor and turbine loss mechanisms that do not alter qualitatively the engine

dynamical behavior are not considered. This, however, comes without a loss in generality. These simplifications can be easily removed if more precision is required.

The international standard for engine station numbering (SAE INTERNATIONAL, 2011) is used throughout this text. It is summarized in Figure 1 for easy reference.

The rest of this thesis is organized as follows:

**Chapter 2** is a literature review of the state of the art in engine and component modeling. It describes the basic elements of compressors, turbines and engines; instabilities in engine systems; and the numerical methods employed to bring these models to life. It is kept relatively brief and plenty of references are provided.

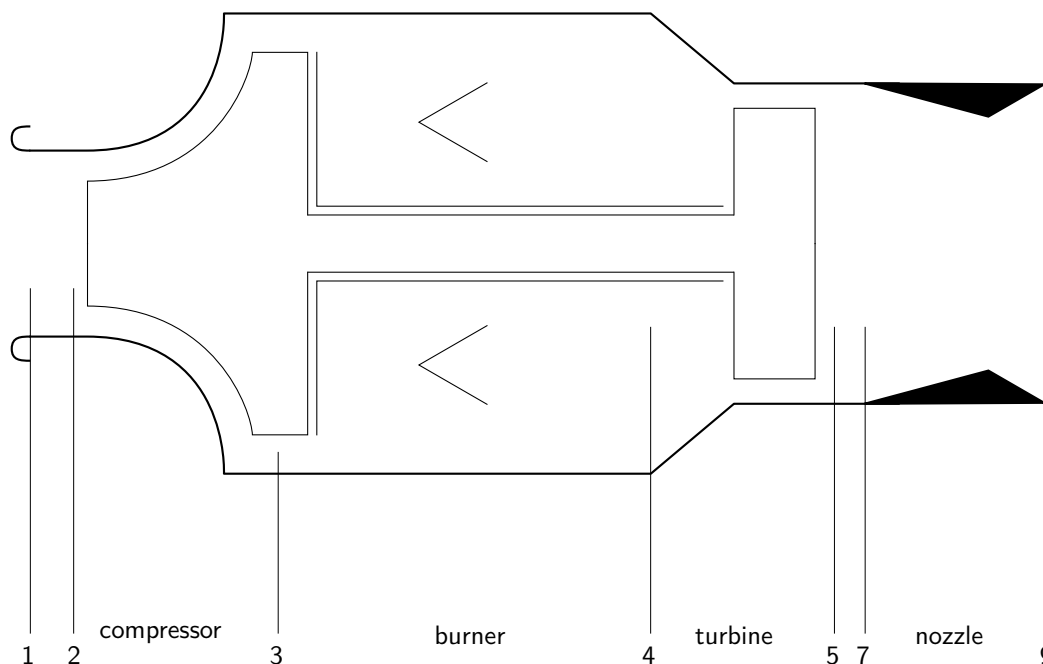
**Chapter 3** develops the methods for simulating a gas turbine engine and its components. It contains tables summarizing the nonlinear system associated with each model and comments on the specifics of the numerical methods used to simulate each aspect of the models. In the end, the engine used for calculations in Chapter 4 is described.

**Chapter 4** presents the results of the simulation, i.e. the component maps, operating line, etc.

**Chapter 5** presents the conclusion and final remarks of this work.

The source code used to produce the results of this work and this text itself is available online (BAHIA MONTEIRO, 2019).

Figure 1 – Schematic diagram of the modeled engine



Source: author's figure.

## 2 Literature review

*To fight and conquer in all our battles is not  
supreme excellence; supreme excellence consists  
in breaking the enemy's resistance without  
fighting.*

-Sun Tzu

The simulation of a gas turbine engine is fundamentally a multidisciplinary matter. It encompass not only the fields of turbomachinery and compressible flow, but also those of dynamical systems and numerical analysis. This chapter is a literature review on these fundamental fields with the goal of providing subsidy for the models derivations in Chapter 3.

Since only the basic equations of compressible flow will be used, they will not be reproduced here. The reader should refer to Anderson (2003) and Shapiro (1953), or simply check the very nice review table available at NASA's website (HALL, 2019).

### 2.1 Euler's equation

Euler's equation arises from the application of the laws of conservation of angular momentum and energy to a turbomachine. It is the cornerstone of performance calculation for compressor and turbines, and is derived in any textbook on turbomachinery (LAKSHMINARAYANA, 1996; DIXON, 1998; SCHOBELI, 2004; HILL; PETERSON, 1991; LOGAN, 2003; BASKHARONE, 2006). Due to its importance, this equation will be briefly derived here.

From the conservation of angular momentum, the external torque to a control volume including a rotating element is given by

$$\vec{\tau} = \dot{m}(\vec{r}_e \times \vec{V}_e - \vec{r}_i \times \vec{V}_i) \quad (2.1)$$

where  $\vec{r}_e \times \vec{V}_e$  is the angular momentum at the control volume's exit,  $\vec{r}_i \times \vec{V}_i$  at the inlet, and  $\dot{m}$  is the mass flow.

Multiplying both sides by the rotational speed ( $\omega$ ),

$$\text{power} = \vec{\omega} \cdot \vec{\tau} = \dot{m}\vec{\omega} \cdot (\vec{r}_e \times \vec{V}_e - \vec{r}_i \times \vec{V}_i) \quad (2.2)$$

Assuming that the system is adiabatic and using the first law of thermodynamics, the specific stagnation enthalpy change is given by

$$\Delta h_0 = \frac{\text{power}}{\dot{m}} = \vec{\omega} \cdot (\vec{r}_e \times \vec{V}_e - \vec{r}_i \times \vec{V}_i) \quad (2.3)$$

$$\Delta h_0 = \omega(r_e V_{ue} - r_i V_{ui}) \quad (2.4)$$

$$\Delta h_0 = U_e V_{ue} - U_i V_{ui} \quad (2.5)$$

where the subscript  $u$  denotes the tangential component of the velocity and  $U$  is the rotor speed at its mean square radius. Furthermore,  $\vec{V}$  denotes the absolute velocities and  $\vec{W}$  the relative



velocities, so that

$$\vec{V} = \vec{U} + \vec{W} \quad (2.6)$$

where  $\vec{U}$  is the rotation speed at the mean flow line.

### 2.1.1 Euler's equation for a centrifugal compressor

For the analysis of a centrifugal compressor, it is useful to write the absolute tangential velocity at the compressor exit ( $V_{u3}$ ) as a function of the radial velocity ( $W_{r3} = V_{r3}$ ), since the latter is more closely related to the mass flow. In fact,  $\dot{m} = \rho W_{r3} A_3$ , where  $A_3$  is the ring shaped exit area of the compressor.

From the velocity triangles in Figure 2, and assuming that the flow exits tangential to the blade surface,

$$V_{u3} = U_3 - W_{r3} \tan \beta_3 \quad (2.7)$$

In a real impeller, the flow does not leave the blade perfectly tangential to its surface. This is accounted by the introduction of a slip factor ( $\sigma$ ) (WIESNER, 1967; AUNGIER, R. H., 1995). The corrected equation for the absolute tangential velocity is then

$$\boxed{V_{u3} = \sigma(U_3 - W_{r3} \tan \beta_3)} \quad (2.8)$$

Substituting Equation (2.8) in Equation (2.5) and dividing both sides by  $U_3^2$ , we have

$$\psi_c = \sigma(1 - \phi_3 \tan \beta_3) - \frac{U_2 V_{u2}}{U_3^2} \quad (2.9)$$

where  $\phi_3 \triangleq \frac{\dot{m}}{A_3 \rho_3 U_3} = \frac{W_{r3}}{U_3}$  is the flow parameter at the compressor exit and  $\psi_c \triangleq \frac{\Delta h_0}{U_3^2}$  is the compressor power coefficient.

The second term accounts for prewhirl in the flow entering the compressor, e.g due to inlet guide vane (IGV). In the case of no prewhirl,  $V_{u2} = 0$ , thus

$$\boxed{\psi_c = \sigma(1 - \phi_3 \tan \beta_3)} \quad (2.10)$$

### 2.1.2 Euler's equation for an axial turbine

Applying Equation (2.5) to an axial turbine stage (Figure 3), we have

$$\Delta h_0 = U_5 V_{u5} - U_4 V_{u4} \quad (2.11)$$

The tangential fluid velocities can be expressed in terms of their axial components, i.e.

$$V_{4u} = V_{4x} \tan \alpha_4 \quad (2.12)$$

$$V_{5u} = U_5 + V_{5x} \tan \beta_5 \quad (2.13)$$

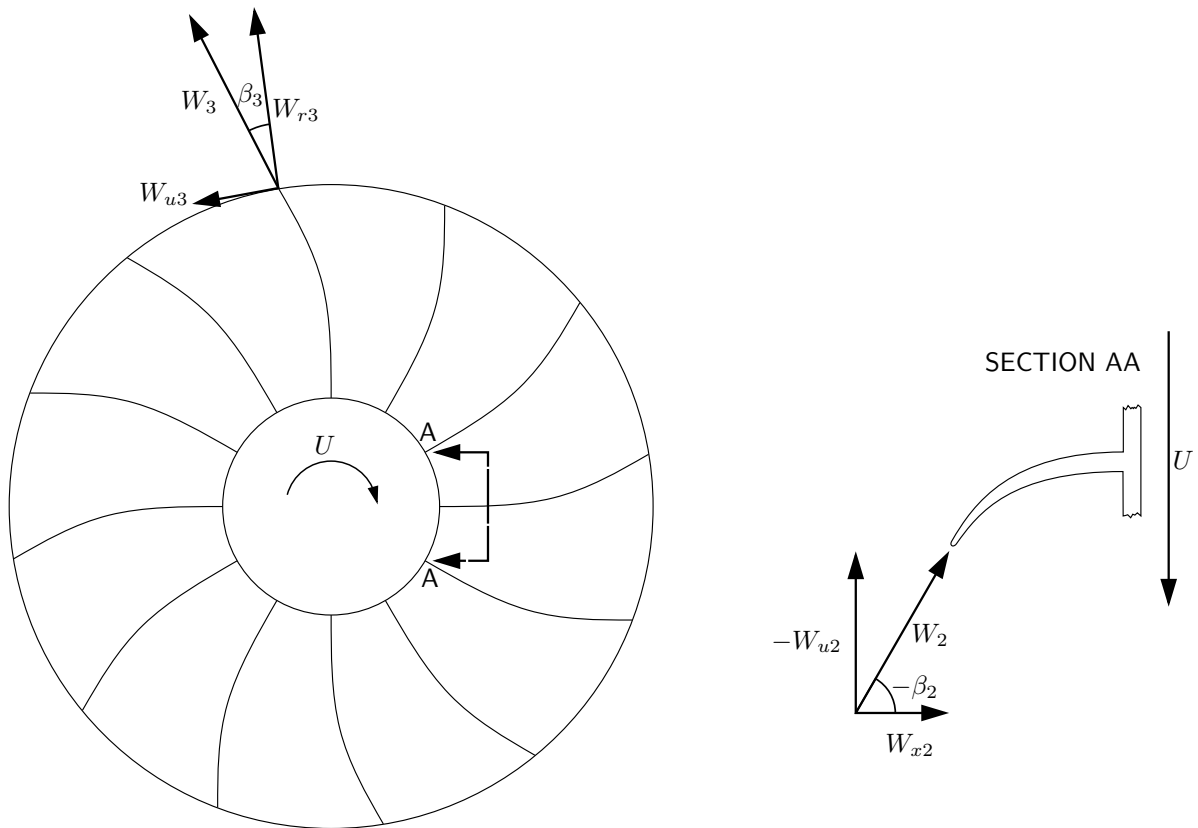
Then,

$$\Delta h_0 = U_5^2 + U_5 V_{5x} \tan \beta_5 - U_4 V_{4x} \tan \alpha_4 \quad (2.14)$$

And finally, making the equation nondimensional by diving both sides by the blade speed at the exit  $U_5$ , we get

$$\boxed{\psi = 1 + \phi_5 \tan \beta_5 - \left(\frac{r_4}{r_5}\right)^2 \phi_4 \tan \alpha_4} \quad (2.15)$$

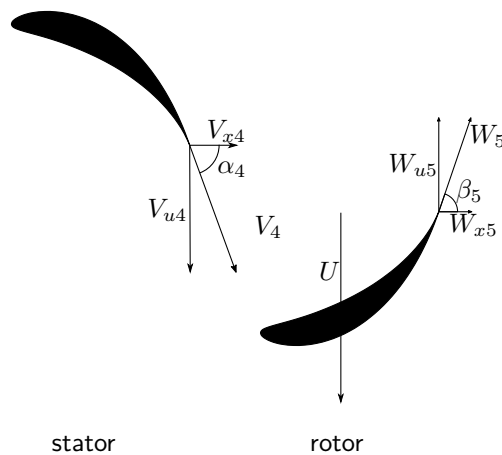
Figure 2 – Velocity triangles for a centrifugal impeller and inducer



Source: author's figure.

The velocities ( $W$ ) relative to the impeller exit (left) and inducer inlet (right) are shown.  $U$  is the impeller tip velocity. The positive sign is defined in the direction of  $U$  for the inlet and opposite to it at the exit, for historical reasons.

Figure 3 – Velocities in an axial turbine stage



Source: author's figure.

where  $\phi_4 = \frac{V_{4x}}{U_5}$  and  $\phi_5 = \frac{V_{5x}}{U_5}$  are the flow parameters and  $\psi = \frac{\Delta h_0}{U_5^2}$  is the power coefficient for the turbine.

## 2.2 Loss models for centrifugal compressors

Euler's equation is only valid for isentropic flow. For a real flow, it must be corrected to account for the generation of entropy in many parts of the compression process. These so called "losses" are modeled in literature as enthalpy corrections for Euler's equation considering both the *actual* process and the *isentropic* process that would generate the same increase in pressure as the former. Each of these losses can be categorized as either *internal* or *parasitic*. Internal losses are those that happen inside the fluid main flow, and result in a decrease in the resulting pressure ratio. Parasitic losses are the ones who happen outside the fluid main flow, and result in a increase in the temperature ratio through the compressor (GALVAS, 1973). Using these definitions, the actual and isentropic enthalpy changes through the compressor are calculate as follows

$$\Delta h_{0\text{actual}} = \Delta h_{0\text{euler}} + \sum_{\text{parasitic losses}} \Delta h_0 \quad (2.16)$$

$$\Delta h_{0\text{isen}} = \Delta h_{0\text{euler}} - \sum_{\text{internal losses}} \Delta h_0 \quad (2.17)$$

The temperature and pressure ratios are given by

$$\frac{T_{03}}{T_{02}} = 1 + \frac{\Delta h_{0\text{actual}}}{c_p T_{02}} \quad (2.18)$$

$$\frac{P_{03}}{P_{02}} = \left( 1 + \frac{\Delta h_{0\text{isen}}}{c_p T_{02}} \right)^{\frac{\gamma}{\gamma-1}} \quad (2.19)$$

and the isentropic and polytropic efficiencies can then be readily calculated as

$$\eta_t = \frac{\Delta h_{\text{isen}}}{\Delta h_{0\text{actual}}} \quad (2.20)$$

$$\eta_p = \frac{\gamma - 1}{\gamma} \left( \frac{\log \frac{P_{02}}{P_{01}}}{\log \frac{T_{02}}{T_{01}}} \right) \quad (2.21)$$

R. H. Aungier (1995) criticizes this calculation because the losses should be regarded as changes in entropy, and not in enthalpy. Despite this shortcoming, the enthalpy approach is widely used.

The losses generally accounted for are listed in Table 1. According to Gravdahl, Frank Willems, et al. (2004), Gravdahl and Egeland (1999) and N. Watson and Janota (1982), the most important losses for stability considerations are the incidence losses and the skin friction losses. The former is destabilizing because it introduces the positive slope of the left hand side of the compressor map. The fluid friction losses are stabilizing because they increase with mass flow, therefore contributing to the decrease in the derivative of the iso-speed lines.

Semi-empirical models are available to predict each individual loss based on compressor geometry and flow conditions. In particular, Oh, Yoon, and Chung (1997) did a review of various

Table 1 – Loss mechanisms for a centrifugal compressor

| <b>Internal</b>                                                                                                                                                                                                                                   | <b>Parasitic</b>                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Incidence loss (in the impeller and dif-fuser)</li> <li>• Blade loading loss</li> <li>• Skin friction loss</li> <li>• Clearance loss</li> <li>• Mixing loss</li> <li>• Vaneless diffuser loss</li> </ul> | <ul style="list-style-type: none"> <li>• Disc friction loss</li> <li>• Recirculation loss</li> <li>• Leakage loss</li> </ul> |

Source: (OH; YOON; CHUNG, 1997).

loss models for each loss mechanism and selected an optimal set when compared to experimental data of three Eckardt impellers and one KIMM impeller. A detailed discussion of each loss mechanism is beyond the scope of this work; the interested reader should refer to Cumpsty (2004).

## 2.3 Loss models for axial turbines

Most of the empirical loss correlations for axial turbines in use today stem from the seminal work by Ainley and Mathieson (1951). This work identified the main loss mechanisms for an axial turbine stage as

**profile loss** due to skin friction and flow separation under high incidence angles of the blades

**annulus loss** due to boundary layer growth on the inner and outer walls of the turbine annulus

**secondary flow loss** due to the tridimensional nature of the flow

**tip clearance loss** due to the vortex formed at the tip of the turbine blades

Several improvements were made to Ainley and Mathieson's model by Mukhtarov and Krichakin (1969), Dunham and Came (1970), Kacker and Okapuu (1982), S. H. Moustapha, Kacker, and Tremblay (1989) and Benner, SA Sjolander, and SH Moustapha (1997) among others (PERSSON, 2015). All of these studies were based on empirical data available from previous turbine rig testing, and thus are valid for turbines with similar geometries. A complete loss system is due to Ronald H. Aungier (2006).

These models provide the loss values as a pressure loss coefficient:

$$Y = \frac{P_{04} - P_{05}}{P_{05} - P_5} \quad (2.22)$$

## 2.4 Deviation models for axial turbines

The flow exiting a turbine blade row does not follow the blade camber line exactly. This is analogous to the slip observed in centrifugal compressors, but needs more elaborate correlations,

thus deserving a section of its own. The simplest correlation available is known as Carter's rule (MATTINGLY, 1996), and states that

$$\delta = \frac{\beta_4 - \beta_5}{4\sqrt{s}} \quad (2.23)$$

where  $\delta$  is the angular deviation between the exit flow and the exit camber angle,  $\beta_4$  and  $\beta_5$  are the inlet and exit airfoil camber angles, and  $s = \frac{\text{chord}}{\text{blade separation}}$  is the solidity.

Another model based on inlet and exit Mach numbers is provided by Ainley and Mathieson (1951), and a more elaborate model involving also blade thickness was proposed by Islam and S. A. Sjolander (1999):

$$\delta = \frac{\frac{\rho_5 V_{x4}}{\rho_5 V_{x5}} s^{-1.1} (\alpha_4 + \beta_5)^{2.25}}{\xi^{1.45} (t_{\max}/c) (22 + 0.22\beta_4^{1.64})} \quad (2.24)$$

where  $\xi$  is the blade geometric incidence (stagger angle),  $t_{\max}/c$  is the blade maximum thickness to chord ratio and all angles are given in degrees.

## 2.5 Gas turbine engine dynamics

A simple engine dynamical model is due to D. A. Fink, Cumpsty, and Greitzer (1992). This model was validated experimentally by Gravdahl, Frank Willems, et al. (2004). It is a system of three ordinary differential equations (ODEs).

$$\dot{\rho}_{04} = \frac{1}{V_p} (\dot{m}_c - \dot{m}_t) \quad (2.25)$$

$$\ddot{m}_c = \frac{A_p}{L_p} (P_{03} - P_{04}) \quad (2.26)$$

$$\dot{\omega} = \frac{1}{I} (\tau_t - \tau_c) \quad (2.27)$$

where  $V_p$  is the combustion chamber volume,  $A_p$  is its cross sectional area and  $L_p$  is its characteristic length.

These equations represent, respectively, the conservation of mass and momentum in the burner and the conservation of angular momentum in the spool.

There are many readily available computer programs that implement this kind of simulation. From these, the most famous is probably the commercial program GasTurb (KURZKE, 2017), followed by the Gas Turbine Simulation Program (GSP) (VISSER; BROOMHEAD, 2000). A comparison of both is presented in Sung (2013). In particular, GasTurb has already been used to simulate model gas turbines (GAO; HUANG, Y., 2011). An open-source alternative is the Toolbox for the Modeling and Analysis of Thermodynamic Systems (T-MATS), from NASA (CHAPMAN et al., 2014). T-MATS provides a library of turbo machinery blocks for use in Simulink. Each of these blocks is actually a wrapper for a function written in C that simulates the component's behavior.

The downside of all these programs is that they rely on component map data provided by the user and do not consider the dynamic of the flow in the burner. The generation of the maps is a substantial effort that must be carried independently, either by scaling maps from similar designs, obtaining the maps experimentally or using semi-empirical methods as described in this

work. On the other hand, disregarding the dynamical behavior of the combustion chamber makes the model unable to predict surge. The surge can still be included through a surge line in the compressor map but, depending on how the map was obtained, this line may not be accurate because the surge depends on the components downstream from the compressor, as discussed in Section 2.6.

## 2.6 Engine stability: surge

There are two main threats to the stability of a GTE: surge and flame extinction. Flame extinction happens when the fuel to air ratio and combustion chamber temperature leave a certain stability envelope (MATTINGLY, 1996), so it does not need to be accounted for directly in an engine model. Thus it will not be covered here. Surge is a dynamical instability in a compressor system caused by its interaction with the components downstream (SPARKS, 1983). It is characterized by periodic backflow accompanied by vibration, a marked dip in thrust and rotation speed, and a characteristic sound, which can be a series of bangs for engines with large internal volume or a low frequency hum for smaller engines. In extreme cases, it can cause damage to the engine. Normally the system can not recover from surge on its own, and either a reduction in fuel flow or the opening of bleed valves is required to return the engine to its steady state operation. (WILLEMS, F.; DE JAGER, 1999).

The surge phenomenon arises naturally from the dynamical model of a compressor system if density transients are taken into account, such as the one from Section 2.5 (GREITZER, 1976a), and has good agreement with experimental data (GREITZER, 1976b; GRAVDAHL; WILLEMS, Frank, et al., 2004). It is nevertheless good to have a criterion for the onset of surge, if only to draw the surge line on the compressor map. A simple model for surge with all the fundamental elements is due to David Allan Fink (1988). This model includes a compressor, a plenum of known volume, some plumbing of known length and an adjustable throttle valve. In an actual engine, the plenum and plumbing are analogous to the combustion chamber, while the throttle can be considered as a simplified turbine-nozzle assembly. The stability criterion derived from this model by linear control theory is

$$C' < \frac{1}{B^2 T'} \quad (2.28)$$

and

$$C' < T' \quad (2.29)$$

where  $C' = \frac{\partial P_{03}}{\partial \phi_c}$  is the derivative of the compressor characteristic,  $T' = \frac{\partial P_{04}}{\partial \phi_t}$  is the derivative of the throttle characteristic and

$$B = \frac{U}{2a_p} \sqrt{\frac{V_p}{A_p L_p}} \quad (2.30)$$

where  $V_p$ ,  $A_p$  and  $L_p$  are the volume, area and characteristic length of the piping system downstream from the compressor.

This model provides a good physical explanation of the mechanism of surge. First, one should notice that, since  $T'$  is always positive, surge can only occur when the compressor characteristic

has a positive derivative, i.e. when an increase in mass flow is accompanied by an increase in the pressure ratio supplied by the compressor. Equation (2.29) shows that if this increase in pressure ratio does not lead to an equal or greater increase in mass flow through the throttle, the increase in compressor mass flow ratio will increase the plenum pressure above that which the compressor can supply, and this will result in backflow. In the negative part of the compressor characteristic, it will be able again to supply the demanded pressure, now made smaller due to backflow. Since this condition will lead to an ever decrease in the plenum pressure, the flow will eventually invert its direction again, and this results in the periodic behavior of surge.

Conversely, Equation (2.28) represents the destabilizing effect of  $B$  in the system. For large values of  $B$ , that is, large burner volumes, the surge limit approaches the point where the compressor characteristic has zero derivative. Since the plenum is very big, any change in the mass flow of the throttle will make a very small difference in the plenum pressure, and so any simultaneous increase in compressor mass flow and pressure ratio will lead to instability. This is the conservative limit for the surge line. Inversely, for small values of  $B$ , that is, small burner volumes, any increase in pressure will lead to a fast increase in mass flow through the throttle, which will instantaneously reduce the pressure, thus having an stabilizing effect. In this case, the onset of surge will be determined by Equation (2.29).

Another result from this model is that while surge is often related to compressor stall, this is not necessarily the case. The surge line for a compressor system can be moved by changing components downstream from it, and operation of compressors beyond the surge line, i.e. with flow angles that would otherwise cause surge, using control systems based on bleed valves is widely reported in the literature (e.g. LIAW; SONG; HUANG, J.-T., 2004), which not be possible if surge was simply an aerodynamical effect.

## 2.7 Nondimensional parameters

Using the Buckingham II Theorem (BUCKINGHAM, 1914), the parameters which govern engine and component performance can be grouped into new, nondimensional parameters. This has the advantage that the engine behavior becomes invariant by physical size, and so the behavior of a small model engine resembles that of a large commercial gas turbine. It is specially beneficial for choosing initial guesses for numerical solvers.

Walsh and Fletcher (2004, chapter 4) provide some very comprehensive tables for the nondimensional groups used for both individual components and complete engines. Componentwise, the relevant parameters are the specific heat ratio, the mass flow parameter at inlet and exit, the blade stagnation Mach number, the stagnation temperature ratio, and the stagnation pressure

ratio. These quantities are defined respectively as

$$\gamma \triangleq \frac{c_p}{c_v} \quad (2.31)$$

$$\text{MFP}_i \triangleq \frac{\dot{m}\sqrt{RT_{0i}}}{A_i P_{0i} \sqrt{\gamma}} = \frac{\dot{m}}{A_i \rho_{0i} a_{0i}} \quad (2.32)$$

$$\text{MFP}_e \triangleq \frac{\dot{m}\sqrt{RT_{0e}}}{A_e P_{0e} \sqrt{\gamma}} \quad (2.33)$$

$$M_b \triangleq \frac{U}{\sqrt{\gamma RT_{0i}}} \quad (2.34)$$

$$\text{stagnation temperature ratio} \triangleq \frac{T_{0e}}{T_{0i}} \quad (2.35)$$

$$\text{stagnation pressure ratio} \triangleq \frac{P_{0e}}{P_{0i}} \quad (2.36)$$

The temperature ratio can be equivalently replaced by the adiabatic or polytropic efficiency, see Section 2.2.

In particular, the mass flow parameter is a function of the Mach number and the specific heat ratio, i.e.

$$\text{MFP}(M, \gamma) = M \left( 1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma+1}{2(\gamma-1)}} \quad (2.37)$$

While there is no closed form inverse for this function, it is invertible if the flow is assumed either subsonic or supersonic and the inverse can be computed numerically, e.g. by a newton scheme provided that the zero in the derivative at  $M = 1$  is removed (DER JR; STAUFFER, 1974).

Although the Reynolds number also has an impact on component performance, its effects are of second order (WALSH; FLETCHER, 2004). For the purpose of this work, it will only be eventually used in the computation of loss models.

For the entire engine, the relevant nondimensional groups are the stationwise mass flow parameter, nondimensional temperature and nondimensional pressure; the reference blade stagnation Mach number and the nondimensional fuel mass flow. The stationwise mass flow parameter is defined in the same way as in Equation (2.32), i.e. for station  $n$ ,  $\text{MFP}_n = \frac{\dot{m}\sqrt{RT_{0n}}}{A_n P_{0n} \sqrt{\gamma}}$ . The nondimensional pressures and temperatures are the local stagnation property normalize by the ambient value. The reference blade stagnation Mach number is taken as the compressor's and the nondimensional fuel flow parameter is defined as  $\frac{\dot{m}_f \sqrt{RT_{01}}}{A_1 P_{01} \sqrt{\gamma}}$ .

## 2.8 Nonlinear systems of differential algebraic equations

As will be shown in Chapter 3, the modeling of a gas turbine engine leads naturally to nonlinear systems of equations of the type

$$f(x) = 0, \quad f : \mathbb{R}^n \rightarrow \mathbb{R}^n \quad (2.38)$$

for component maps and static engine operating points, and to differential-algebraic equation systems of the type

$$f(t, x, \dot{x}) = 0, \quad f : \mathbb{R}^{2n+1} \rightarrow \mathbb{R}^n \quad (2.39)$$

for the dynamical engine model. This particular formulation is known in the literature as a differential-algebraic equation (DAE) system (BRENAN; CAMPBELL; PETZOLD, 1995). The surge



model, on the other hand, is best expressed as a partial differential-algebraic equation system of the form

$$f(x, \nabla x) = 0, \quad f : \mathbb{R}^{n^2+n} \rightarrow \mathbb{R}^n \quad (2.40)$$

where  $\nabla x$  is the Jacobian of  $x$ . However, the literature on this type of system is sparse at best and no widely used software package could be found. This section will therefore focus itself on the problems which can be expressed in the form of Equations (2.38) and (2.39).

Walsh and Fletcher (2004) suggests that one way of solving theses systems is solving for one variable at a time in an approach known as successive loop closure or serial nested loop. For example, in the case of an engine's static model, Walsh and Fletcher (2004) proposes that, starting with a fixed shaft speed, one first guesses the operating point on the compressor map and the turbine inlet temperature  $T_{04}$ . Then,  $T_{04}$  is adjusted until the mass flow predicted by the turbine map matches the guess for the turbine map. The nozzle performance is then evaluated and the position on the compressor map is changed to match the nozzle mass flow ratio. This procedure is repeated until convergence of both parameters is obtained.

This method is severely flawed. As remarked by Walsh and Fletcher (2004), it becomes very inefficient when the number of guesses is more than approximately five. Furthermore, this technique is not only hard to set up and dependent on intuition acquired through experience, but it is also very prone to diverge (sometimes wildly so) if the updated values for the variables are not chosen just right. A more robust approach is to use proven solvers for non-linear systems, referred by Walsh and Fletcher (2004) as the "matrix" solution method. This method consists on using gradient based algorithms to solve the system simultaneously for all variables.

There are two main approaches to solving non-linear systems numerically in wide use today. One is to use classical root finding algorithms based on optimization theory, such as Levenberg-Marquardt and Powell's hybrid method (MORE; GARROW; HILLSTROM, 1980). The other is to use path following methods, such as DAE system solvers and homotopy methods (ALLGOWER; GEORG, 1997; RABIER; RHEINOLDT, 2002; WATSON, L. T. et al., 1997; HINDMARSH et al., 2005).

It should be noted that every system expressed as a nonlinear system can also be represented as a DAE. This is most useful for under determined systems where not only one solution is sought but a unidimensionally parameterized continuum of solutions, such as an operating line, choke line, surge line, etc.

### 3 Methods

*It's not denial.*

*I'm just very selective about the reality I accept.*

*-Calvin*

This section deals with the models used for the engine compressor, turbine and nozzle, as well as the methods used for calculating the engine steady operation (engine matching), surge, dynamical behavior and performance parameters. Throughout this section, the ideal gas assumption is widely used, as is the assumption that the performance of a component can be separated in an ideal performance and a loss model. In the end, the engine from which the parameters for the model were extracted is described.

The component models are normally referred to as component maps or characteristic lines. A component map or characteristic is, in a general sense, the functional relationship between a component's mass flow rate, rotation speed, pressure ratio and efficiency. These five unknowns are linked by two constraints, namely Euler's equation and a loss model. This leaves two free variables, i.e. a two dimensional set of solutions. Usually compressor maps are plotted using the mass flow rate as the abscissa and the pressure ratio as the ordinate, with contours for constant rotation speed and efficiency. For turbines, the rotational speed and efficiency is not uniquely defined for a combination of mass flow rate and pressure ratio. This led to the use of the artificial parameter of rotational speed times mass flow rate for the abscissa.

The name characteristic is most often used in reference to the iso-speed lines of a turbine or compressor when plotted in the plane of flow parameter vs. pressure ratio. For incompressible flow this leads to the collapse of the constant rotational speed lines, thus being very convenient.

#### 3.1 Compressor model

To write Euler's equation 2.5 in terms of the nondimensional parameters introduced in Section 2.7, it suffices to rewrite  $\phi_3$  and  $\psi$  as a function of them. Thus

$$\begin{aligned}\phi_3 &= \frac{W_{2r}}{U_3} = \frac{\frac{\dot{m}}{\rho_3 A_2}}{M_b a_{02}} = \frac{\frac{\dot{m}}{A_2 \rho_{02} a_{02}}}{M_b} \frac{a_{03}}{a_{02}} \left(1 + \frac{\gamma - 1}{2} M_3^2\right)^{\frac{1}{\gamma-1}} \\ &= \frac{\text{MFP}_3}{M_b} \sqrt{\frac{T_{03}}{T_{01}}} \left(1 + \frac{\gamma - 1}{2} M_3^2\right)^{\frac{1}{\gamma-1}}\end{aligned}\quad (3.9)$$

The value of  $M_3$  can be obtained from  $\text{MFP}_3$  using Equation (2.37), while the latter is given by

$$\text{MFP}_3 \triangleq \frac{\dot{m} \sqrt{RT_{02}}}{A_2 P_{02} \sqrt{\gamma}} = \text{MFP} \frac{A_2 P_{02}}{A_3 P_{03}} \sqrt{\frac{T_{03}}{T_{02}}}\quad (3.10)$$

Notice that Equation (3.10) makes no assumption of the process 1–2 being isentropic. Furthermore  $\text{MFP}_3$  can be used along with  $\text{MFP}_2$  to check for choking.

Finally, the load coefficient is given by

$$\psi \triangleq \frac{\Delta h_0}{U_3^2} = \frac{c_p(T_{02} - T_{01})}{U_3^2} = \frac{\frac{c_p(T_{03} - T_{01})}{\gamma R T_{02}}}{M_b^2} = \frac{\frac{T_{03}}{T_{02}} - 1}{M_b^2} \left(\frac{1}{\gamma - 1}\right)\quad (3.11)$$

Table 2 – Compressor model problem statement

Given  $\gamma$  and the compressor geometry  $\mathcal{G} \triangleq$   
 $(\beta_2, \beta_3, \alpha_3, \frac{A_3}{A_2}, \frac{D_{1t}}{D_2}, \frac{D_{1h}}{D_2}, Z)$ ,  
 find  $\text{MFP}_2, \text{MFP}_3, M_b, \frac{P_{03}}{P_{02}}, \frac{T_{03}}{T_{02}}$  which satisfy

$$\text{MFP}_2 - \text{MFP}_3 \frac{A_3}{A_2} \frac{P_{03}}{P_{02}} \sqrt{\frac{T_{02}}{T_{03}}} = 0 \quad (3.1)$$

$$\frac{T_{03}}{T_{02}} - [1 + (\gamma - 1)\psi_{\text{actual}} M_b^2] = 0 \quad (3.2)$$

$$\frac{P_{03}}{P_{02}} - [1 + (\gamma - 1)\psi_{\text{isen}} M_b^2]^{\frac{\gamma}{\gamma-1}} = 0 \quad (3.3)$$

where

$$\psi_{\text{isen}} = \psi_{\text{euler}} - \sum_{\text{internal losses}} \Delta\psi(\psi_{\text{euler}}, \phi_2, \phi_3, \mathcal{G}) \quad (3.4)$$

$$\psi_{\text{actual}} = \psi_{\text{euler}} + \sum_{\text{parasitic losses}} \Delta\psi(\psi_{\text{euler}}, \phi_2, \phi_3, \mathcal{G}) \quad (3.5)$$

$$\psi_{\text{euler}} = \sigma(1 - \phi_3 \tan \beta_3) \quad (3.6)$$

$$\phi_2 = \frac{\text{MFP}_2}{M_b} \left(1 + \frac{\gamma-1}{2} M_2^2\right)^{\frac{1}{\gamma-1}} \quad (3.7)$$

$$\phi_3 = \frac{\text{MFP}_3}{M_b} \left(1 + \frac{\gamma-1}{2} M_3^2\right)^{\frac{1}{\gamma-1}} \sqrt{\frac{T_{03}}{T_{02}}} \quad (3.8)$$

$$\text{MFP} = M \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \quad (2.37)$$

and  $\sigma(\mathcal{G}), \Delta\psi(\psi_{\text{euler}}, \phi_2, \phi_3, \mathcal{G})$  are given by a loss model.

At this point the loss models discussed in Section 2.2 are introduced, and the values of  $\psi_{\text{actual}}$  and  $\psi_{\text{isen}}$  are obtained. The temperature and pressure ratios can then be readily calculated from

$$\frac{T_{03}}{T_{02}} = (\gamma - 1)\psi_{\text{actual}} M_b^2 + 1 \quad (3.12)$$

$$\frac{P_{03}}{P_{02}} = [(\gamma - 1)\psi_{\text{isen}} M_b^2 + 1]^{\frac{\gamma}{\gamma-1}} \quad (3.13)$$

In this work, only the losses due to incidence and skin friction were considered, because they are the ones most important for compressor stability. The model chosen for skin friction was the one selected by the comparative study of Oh, Yoon, and Chung (1997) and the incidence losses were modeled as the complete conversion of all kinetic energy normal to the blade camber line to heat (GALVAS, 1973), i.e.

$$\Delta\psi = \left[\left(\frac{r_3}{r_2}\right)^2 + \phi_2^2\right] \sin(\beta_2 - \beta_{2\text{opt}}) \quad \text{for the inducer} \quad (3.14)$$

$$\Delta\psi = (\psi_{\text{euler}}^2 + \phi_2^2) \sin(\alpha_3 - \alpha_{3\text{opt}}) \quad \text{for the vanned diffuser} \quad (3.15)$$

where  $\beta_2$  and  $\alpha_3$  are the actual blade (metal) angles and  $\beta_{2\text{opt}}$  and  $\alpha_{3\text{opt}}$  are the flow angles for each flow condition.

Table 3 – Turbine model problem statement

---

Given  $\gamma$  and the turbine geometry  $\mathcal{G} \triangleq (\alpha_4, \beta_5, \frac{A_5}{A_4}, \frac{D_5}{D_4})$ ,  
find  $\text{MFP}_4, \text{MFP}_5, M_b, \frac{P_{05}}{P_{04}}, \frac{T_{05}}{T_{04}}$  which satisfy

$$\text{MFP}_4 - \text{MFP}_5 \frac{A_5}{A_4} \frac{P_{05}}{P_{04}} \sqrt{\frac{T_{04}}{T_{05}}} = 0 \quad (3.16)$$

$$\frac{T_{05}}{T_{04}} - [1 + (\gamma - 1)\psi_{\text{euler}} M_b^2] = 0 \quad (3.17)$$

$$\frac{P_{05}}{P_{04}} - [1 + (\gamma - 1)\psi_{\text{euler}} M_b^2]^{\frac{\gamma}{\gamma-1}} = 0 \quad (3.18)$$

where

$$\psi_{\text{euler}} = 1 + \phi_5 \tan \beta_5 - \left(\frac{D_4}{D_5}\right)^2 \phi_4 \tan \alpha_4 \quad (3.19)$$

$$\phi_4 = \frac{\text{MFP}_4}{M_b} \left(1 + \frac{\gamma-1}{2} M_4^2\right)^{\frac{1}{\gamma-1}} \quad (3.20)$$

$$\phi_5 = \frac{\text{MFP}_5}{M_b} \left(1 + \frac{\gamma-1}{2} M_5^2\right)^{\frac{1}{\gamma-1}} \sqrt{\frac{T_{05}}{T_{04}}} \quad (3.21)$$

$$\text{MFP} = M \left(1 + \frac{\gamma-1}{2} M^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}} \quad (2.37)$$


---

This is a more physical model than the one suggested by Oh et al., and captures better the loss variation with incidence, since Oh et al's model does not include flow angles.

The compressor model is summarized in Table 2.

## 3.2 Turbine model

The turbine model is summarized in Table 3. Its development is mostly analogous to that of the compressor, with the exception of the loss and deviation models. Since the turbine does not exhibit flow instability phenomena such as compressor surge, which depends heavily on the incidence losses, the turbine was assumed isentropic. Nevertheless, a loss model based on pressure loss coefficients  $Y_p, Y_s, Y_k$ , etc. could easily be included in this model by modifying Equation (3.18) in the following way

$$\frac{P_{05}}{P_{04}} - \frac{[1 + (\gamma - 1)\psi_{\text{euler}} M_b^2]^{\frac{\gamma}{\gamma-1}}}{1 + Y \left\{ 1 - \left[ 1 + \frac{\gamma-1}{2} M_5^2 \right]^{\frac{\gamma-1}{\gamma}} \right\}} = 0 \quad (3.22)$$

where

$$Y = Y_p + Y_s + Y_k \quad (3.23)$$

This correction is derived as follows. The pressure loss coefficient is given by

$$Y = \frac{P_{0i} - P_{0e}}{P_{0e} - P_e} \quad (3.24)$$

Now we divide the numerator and the denominator by  $P_{0e}$  and isolate the pressure ratio term

$$\frac{P_{0i}}{P_{0e}} = \frac{1}{1 + Y \left\{ 1 - \left[ 1 + \frac{\gamma-1}{2} M_e^2 \right]^{-\frac{\gamma-1}{\gamma}} \right\}} \quad (3.25)$$

This pressure ratio loss term must be multiplied to the ideal (isentropic) pressure ratio given by the second term of Equation (3.18). If more than one blade row is considered (e.g. one for the stator and another one for the rotor) the pressure ratio losses ( $Y$ ) for each must be joined by multiplication.

### 3.3 Nozzle model

The nozzle model must consider the both unchoked and choked flow regimes. Under unchoked flow, the nozzle static exit pressure ( $P_9$ ) must be equal to the ambient pressure ( $P_a$ ). When the nozzle is choked, the exit velocity must be sonic and the exit pressure must be greater than the ambient pressure. In order to be useful for the non-linear engine model, these conditions must be expressed as a single, preferably differentiable, equation.

Starting by the unchoked case, the exit Mach number is given by

$$M_9 = \sqrt{\frac{2}{\gamma-1} \left[ \left( \frac{P_{05}}{P_9} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} \quad (3.26)$$

where  $P_9 = P_a$ , and the choke pressure ratio is given by

$$\left( \frac{P_{05}}{P_9} \right)^* = \left( \frac{\gamma+1}{2} \right)^{\frac{\gamma}{\gamma-1}} \quad (3.27)$$

For pressure ratios greater than the choke pressure ratio, the Mach number does not increase above unity, since the nozzle is convergent only. Therefore, the nozzle behavior for either choked or unchoked flow is

$$M_9 \left( \frac{P_{05}}{P_a} \right) = \begin{cases} 0 & \text{if } \frac{P_{09}}{P_9} < 1 \\ \sqrt{\frac{2}{\gamma-1} \left[ \left( \frac{P_{09}}{P_a} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right]} & \text{if } 1 \leq \frac{P_{09}}{P_9} < \left( \frac{\gamma+1}{2} \right)^{\frac{\gamma}{\gamma-1}} \\ 1 & \text{if } \frac{P_{09}}{P_9} \geq \left( \frac{\gamma+1}{2} \right)^{\frac{\gamma}{\gamma-1}} \end{cases} \quad (3.28)$$

This function is continuous but its derivative has discontinuities at the connecting points. Although the case  $\frac{P_{09}}{P_9} < 1$  has no physical meaning (the stagnation pressure can never be less than the static pressure in isentropic flow), it may arise during the iterative solution of the nonlinear system, hence it is useful to have it mathematically well defined. In this case it was simply defined as a constant chosen to ensure continuity. It will be shown that this choice also provides for a continuous derivative in the final function.

The singularity in the derivative at sonic speed can be removed by using the mass flow parameter (MFP) instead of the Mach number, since  $\frac{\partial \text{MFP}}{\partial M} \Big|_{M=1} = 0$  and  $\frac{\partial M_9}{\partial P_{05}/P_a} \Big|_{M_9=1}$  is finite, therefore  $\frac{\partial \text{MFP}_9}{\partial P_{05}/P_a} \Big|_{M_9=1} = 0$ . Assuming the nozzle isentropic, the stagnation conditions at the throat match the ones at the turbine exit, so

$$\text{MFP}_9 = \text{MFP}_5 \frac{A_5}{A_9} \quad (3.29)$$

Using Equation (3.28),

$$\text{MFP}_5 \frac{A_5}{A_9} - \text{MFP}(M_9(\frac{P_{05}}{P_a}), \gamma) = 0 \quad (3.30)$$

where MFP is taken as a function of  $M$  and  $\gamma$  as defined in Equation (2.37).

The final step is to remove the singularity in the derivative of the second term at  $\frac{P_{09}}{P_9} = 1$ . It is easy to verify that multiplying the equation by  $(\frac{P_{09}}{P_9} - 1)$  removes the singularity and the resulting derivative at that point is zero. Therefore, the nozzle constraint is

$$\left( \frac{P_{05}}{P_a} - 1 \right) \text{MFP}_5 \frac{A_5}{A_9} - \left( \frac{P_{05}}{P_a} - 1 \right) \text{MFP} \left( M_9 \left( \frac{P_{05}}{P_a} \right), \gamma \right) = 0 \quad (3.31)$$

This equation holds for all  $\frac{P_{05}}{P_a} \geq 1$ , is mathematically well defined for all real values of  $\text{MFP}_5$  and  $\frac{P_{05}}{P_a}$ , is one time differentiable and has a continuous derivative everywhere.

### 3.4 Engine static model

The engine static model, also known as the matching model, describes the conditions in which the engine can operate indefinitely without changes in throttle input or outside conditions (pressure and temperature).

The engine model used is of the lumped volume type. This means that each component (compressor, burner, turbine and nozzle) is modeled as a single point in space and conservation laws are applied to them. This type of model is also known as zero dimensional (0-D) or as parametric cycle analysis, and is widely used in both industry and academy as a tool for preliminary design and performance analysis.

A schematic of the engine components considered for the simulation of the present gas turbine engine is shown in Figure 1.

The model was derived considering following constraints:

1. conservation of energy through the spool, Equation (3.33);
2. conservation of mass in the burner, Equation (3.34);
3. conservation of energy in the combustion process, Equation (3.35);
4. the nozzle equation, Equation (3.31);
5. and the compressor and turbine models, Tables 2 and 3.

This model is shown in Table 4. It has 12 variables and 11 equations, so it has a one dimensional solution set i.e. the work line.

### 3.5 Engine dynamical model

The dynamical engine model adopted was the one described in Section 2.5. It is similar to the one described in Section 3.4 with the following changes:

1. The conservation of energy in the spool, Equation (3.33), is replaced by Equation (3.37) to allow spool acceleration;

2. The conservation of mass in the burner, Equation (3.34), is replaced by Equation (3.38) to allow transient changes in density within the burner;
3. The implicit assumption that the pressure in the burner is constant from inlet to exit ( $P_{03} = P_{04}$ ) is dropped and replaced by the conservation of momentum in the combustion chamber, Equation (3.39);
4. and the ideal gas equation is added to link the pressure, density and temperature in the combustion chamber.

This results in the nonlinear system shown in Table 5. It has 14 variables and 13 equations, thus it is a single input system, like the static model. For a control problem, the input variable would be the fuel mass flow.

### 3.6 Surge model

Due to the nonavailability of an off the shelf partial DAE solver, the model described in Section 2.6 was not implemented. Instead, the surge line was estimated to be at a MFP equivalent to 80% of the required MFP to achieve minimum angular speed for a given pressure ratio, i.e. 20% before the peak of the compressor characteristic. To implement Section 2.6's model in a partial DAE or with discretized partial derivatives, the static model (Section 3.4) could be used with the nozzle equation, Equation (3.31), removed since for surge it is important to consider the engine at any overall pressure ratio ( $\frac{P_{05}}{P_{02}}$ ). The addition of the stability criteria would maintain the number of equations, thus the solution set would still be a line.

### 3.7 Numerical aspects

As was shown in this section and summarized in ??, the models for a gas turbine engine and its components are expressed naturally in the form of coupled (sometimes differential) non-linear systems of equations. These systems are inherently coupled because their behavior depends on the flow angles on both the inlet and exit, and these can not be calculated independently of one another. As discussed in Section 2.8, these problems are examples of DAE systems, and many of them do not have explicit derivatives in their formulation.

In this section, the numerical approach used to calculate each of the models described in this chapter will be exposed. They consist of a mix of nonlinear solvers, namely Powell's hybrid method and Levenberg-Marquardt, as implemented by MINPACK and available in the Python package *scipy*, and of the newer DAE solver IDA, available as part of SUNDIALS (HINDMARSH et al., 2005) with a Python interface by Assimulo (ANDERSSON; FÜHRER; ÅKESSON, 2015).

#### 3.7.1 Generation of component maps

A classic nonlinear solver was used, coupled with a breadth-first search approach for propagating previous solutions as initial guesses. Firstly, the initial point in the  $x$  and  $y$ -axis is chosen randomly within the choke boundaries, pressure and temperature ratios are unit, Mach numbers are between 0 and 1. If convergence is achieved a breadth-first search considering that each point

in the grid is a graph node connected to the points up, down, left, right and in the diagonals is done. Considering this search tree, the solution for each parent is used as a initial guess for its children. Provided that the grid is thin enough, this provides very good initial guesses for the nonlinear solver, ensuring fast and robust convergence. Furthermore, since each node has 8 neighbours, they can have up to 8 different initial guesses, which makes this approach even more robust.

### 3.7.2 Choke line

Drawing the choke line for the compressor is generally straightforward, since it is practically linear. The first step is to find the point where inlet and exit are simultaneously choked. For pressure ratios above this critical value, only the inlet is choked and the choke line is vertical. For pressure ratios below this value the exit is choked. The choke line was verified to be a straight in this region, so that just solving for a choked point at another pressure ratio, e.g. the minimum pressure ratio of interest (normally unit) is needed to fully define it.

Drawing the line for the turbine is slightly more elaborate, because although it is linear in a plot of MFP vs. pressure ratio it is not in a plot of  $\text{MFP}M_b$  vs. expansion ratio. For the particular turbine used in this work, the choke was found to be determined by the turbine exit, so this was the condition checked. After a initial point on the choke line was found, it was used as a initial guess for finding the next. This approach, however, is not very robust. We recommend instead using a KS aggregation function to join the two choke conditions in one equation, and then use a path following solver to find the choke line (POON; MARTINS, 2007).

### 3.7.3 Generation of the working line

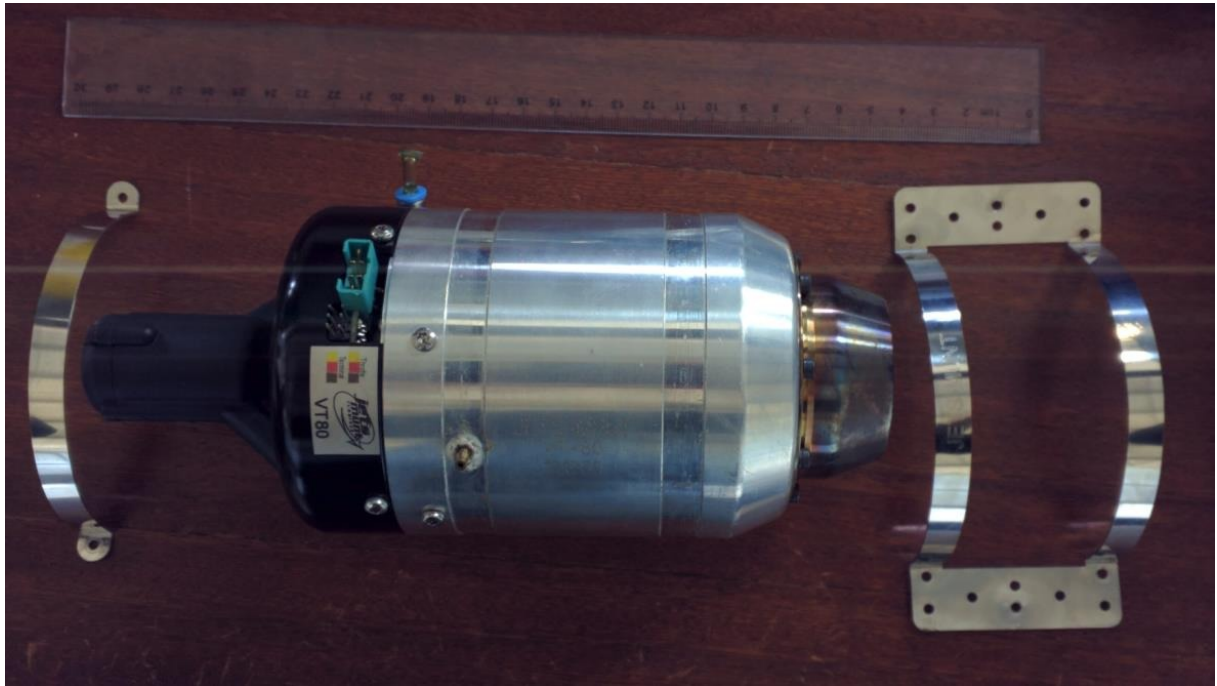
We recommend firstly optimizing the engine model for maximum efficiency, subject to the model constraints. In general high efficiency points are more linear, so convergence is facilitated. After that, use a DAE solver to continue the line forwards and backwards.

## 3.8 The VT-80 engine

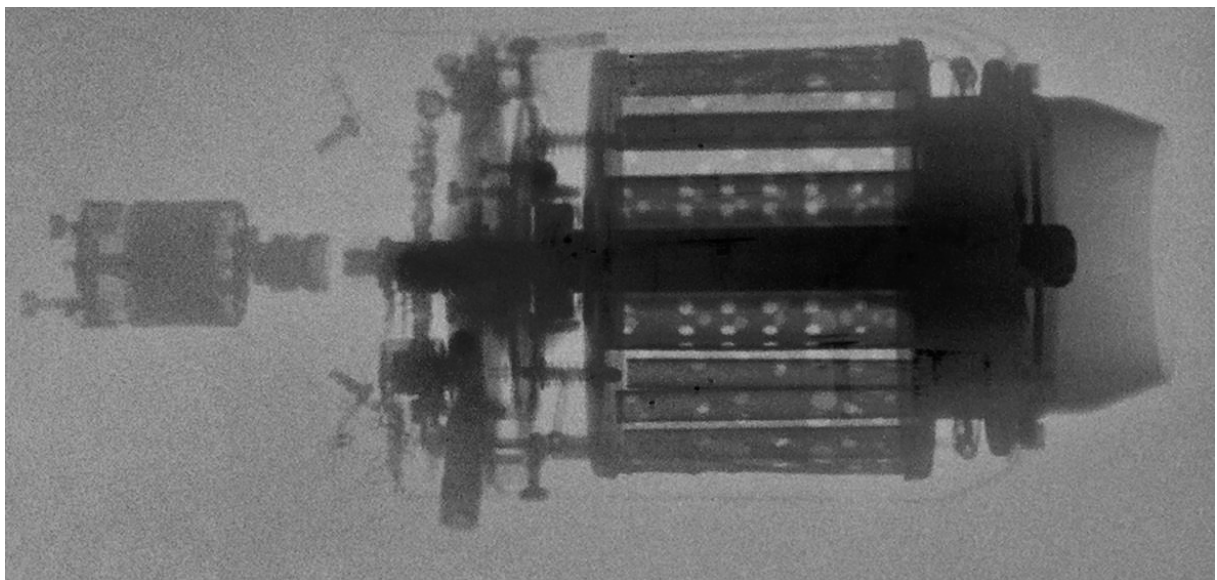
The engine chosen to which apply these models was Jet-Munts VT-80. This engine was developed for hobby use in airplane models. It is a single spool jet engine, with a single stage centrifugal compressor and also single stage axial turbine, as seen in Figure 4. It stems from the original micro turbine desin by Schreckling (1994) and improved by Kamps (1999), among others. It runs on aviation kerosene premixed with oil. The specifications provided by the manufacturer are in Table 7. Oliveira (2017a) estimates the compressor pressure ratio to be around 2.2 based on similar engines. Engines and components drawings with all dimensions relevant for the simulations are annexed to this work. This drawings were based on Kamps' design, measurements of the actual engine, and CAD models found online.



Figure 4 – JetsMunt VT-80 jet engine



(a) Picture with a 30cm ruler for scale



(b) X-ray picture

Source: (OLIVEIRA, 2017a).

Table 4 – Engine steady state model problem statement

Given the constants required as inputs for the compressor and turbine maps (Tables 2 and 3), the flight Mach number ( $M_{\text{flight}}$ ), the free flow stagnation temperature ( $T_{01}$ ) and pressure ( $P_{01}$ ), the fuel lower calorific value (FHV), the combustion efficiency ( $\eta_f$ ), the specific heat of clean air ( $c_{pc}$ ) and combustion products ( $c_{pt}$ ) find  $MFP_2$ ,  $MFP_3$ ,  $MFP_4$ ,  $MFP_5$ ,  $M_{bc}$ ,  $M_{bt}$ ,  $\frac{P_{03}}{P_{02}}$ ,  $\frac{P_{04}}{P_{05}}$ ,  $\frac{T_{03}}{T_{02}}$ ,  $\frac{T_{04}}{T_{05}}$ ,  $T_{04}$ ,  $\dot{m}_f$  which satisfy

$$\omega_c - \omega_t = 0 \quad (3.32)$$

$$\dot{m}_2 c_{pc}(T_{03} - T_{02}) - \dot{m}_4 c_{pt}(T_{04} - T_{05}) = 0 \quad (3.33)$$

$$\dot{m}_3 + \dot{m}_f - \dot{m}_4 = 0 \quad (3.34)$$

$$T_{03} + \frac{\dot{m}_f \eta_f \text{FHV}}{\dot{m}_3 c_{pc}} - T_{04} = 0 \quad (3.35)$$

$$\text{nozzle equation} \quad (\text{Equation (3.31)})$$

$$\text{compressor map} \quad (\text{Table 2})$$

$$\text{turbine map} \quad (\text{Table 3})$$

where the unknowns are dimensionalized as described in the section of nondimensional parameters

Note that although this model is given in its dimensionalized form, it can easily be made non-dimensional if

- Equation (3.35) and  $T_{04}$  are normalized by  $T_{01}$
- Equation (3.34) and  $\dot{m}_f$  are normalized by  $\rho_{01} A_1 \sqrt{\gamma_c R_c T_{01}}$
- Equation (3.33) is normalized by  $c_{pc} T_{01} \rho_{01} A_1 \sqrt{\gamma_c R_c T_{01}}$
- Equation (3.32) is normalized by  $D_1 \sqrt{\gamma_c R_c T_{01}}$

The input parameters for this case are  $\frac{c_{pt}}{c_{pc}}$ ,  $\frac{\eta_f \text{FHV}}{c_{pc} T_{01}}$  and the inputs for the compressor and turbine maps (Tables 2 and 3).

Table 5 – Engine dynamic model problem statement

Given the constants required as inputs for the compressor and turbine maps (Tables 2 and 3), the flight Mach number ( $M_{\text{flight}}$ ), the free flow stagnation temperature ( $T_{01}$ ) and pressure ( $P_{01}$ ), the fuel lower calorific value (FHV), the combustion efficiency ( $\eta_f$ ), the specific heat of clean air ( $c_{pc}$ ) and combustion products ( $c_{pt}$ ) find  $MFP_2$ ,  $MFP_3$ ,  $MFP_4$ ,  $MFP_5$ ,  $M_{bc}$ ,  $M_{bt}$ ,  $\frac{P_{03}}{P_{02}}$ ,  $\frac{P_{04}}{P_{05}}$ ,  $\frac{T_{03}}{T_{02}}$ ,  $\frac{T_{04}}{T_{05}}$ ,  $T_{04}$ ,  $P_{04}$ ,  $\rho_{04}$ ,  $\dot{m}_f$  which satisfy

$$\omega_c - \omega_t = 0 \quad (3.36)$$

$$I\omega_c\dot{\omega}_c - [\dot{m}_2c_{pc}(T_{03} - T_{02}) - \dot{m}_4c_{pt}(T_{04} - T_{05})] = 0 \quad (3.37)$$

$$\dot{\rho}_{04}V_p - (\dot{m}_3 + \dot{m}_f - \dot{m}_4) = 0 \quad (3.38)$$

$$\ddot{m}_c - \frac{A_p}{L_p}(P_{03} - P_{04}) = 0 \quad (3.39)$$

$$T_{03} + \frac{\dot{m}_f\eta_f\text{FHV}}{\dot{m}_3c_{pc}} - T_{04} = 0 \quad (3.40)$$

$$\frac{\rho_{04}R_tT_{04}}{P_{04}} - 1 = 0 \quad (3.41)$$

nozzle equation (Equation (3.31))

compressor map (Table 2)

turbine map (Table 3)

where the unknowns are dimensionalized as described in the section of nondimensional parameters

Nondimensionalizations similar to the ones described in Table 4 can be carried out.

Table 6 – Equation balance for the non-linear models

| model            | variables | equations |     |     | DOF |
|------------------|-----------|-----------|-----|-----|-----|
|                  |           | algebraic | ODE | PDE |     |
| compressor       | 5         | 3         | -   | -   | 2   |
| turbine          | 5         | 3         | -   | -   | 2   |
| engine (static)  | 12        | 11        | -   | -   | 1   |
| engine (dinamic) | 15        | 10        | 3   | -   | 2   |
| surge (static)   | 12        | 10        | -   | 1   | 1   |

Table 7 – VT80 Specifications

|                                    |                   |
|------------------------------------|-------------------|
| Diameter (external)                | 90 mm             |
| Length                             | 240 mm            |
| Weight (bare)                      | 950 g             |
| Weight (with accessories)          | 1075 g            |
| Nominal thrust @ 15C and sea level | 80 N              |
| Idle thrust                        | 4 N               |
| Max rpm                            | 150 000 rpm       |
| Idle rpm                           | 45 000 rpm        |
| Heat soak rpm                      | 12 000 rpm        |
| EGT @ max rpm                      | 600±50 °C         |
| Fuel consumption @ max rpm         | 0.29 L/min        |
| Fuel/oil                           | Kerosene + 4% oil |

Source: (JETS MUNT SL, 2012).

## 4 Results and discussion

*Que tanto de peça para dar defeito.*

*(That's a lot of parts to malfunction.)*

*- My grandfather about my father's new car*

Figure 5 shows the compressor and turbine maps and the work lines for two nozzles: the actual 45mm nozzle and a redesigned 35mm nozzle. The compressor map is very in line with what is expected. Iso-rotation lines peak slightly to the right of the surge line and then fall steeply towards the choke line. Choke occurs in the compressor exit for pressure ratios below approximately 1.4, as shown by the sloped choke line. There is a region of high efficiency near the surge line and efficiencies fall for high mass flows and low pressure ratios. The turbine map, on the other hand, is not what is typically found in the literature. This happens because this turbine chokes at its exit in almost all operating conditions. This leads to the iso-speed lines only becoming vertical near the peak expansion ratio, and the choke line not being horizontal. The turbine was considered isentropic, so there are no iso-efficiency lines on its map.

Normally, well designed turbines choke on their inlets, and stay choked for most of the working range. This helps to extract maximum work from the fluid and also reduces the turbine operating range. In order to achieve an inlet choke with this turbine, its inlet area should be severely reduced.

The nominal working line, with the 45mm nozzle, shows that there is a poor match between compressor, turbine and nozzle. The compressor, in particular, is operating at quite low efficiencies and chokes with a very low value of  $T_{04}$ . Reducing the exit area of the nozzle by 10mm proved an easy way to increase its efficiency and avoid the choking region, while still keeping a good surge margin.

To sum up, following design recommendations can be made based on the analysis of the component maps and work lines in Figure 5:

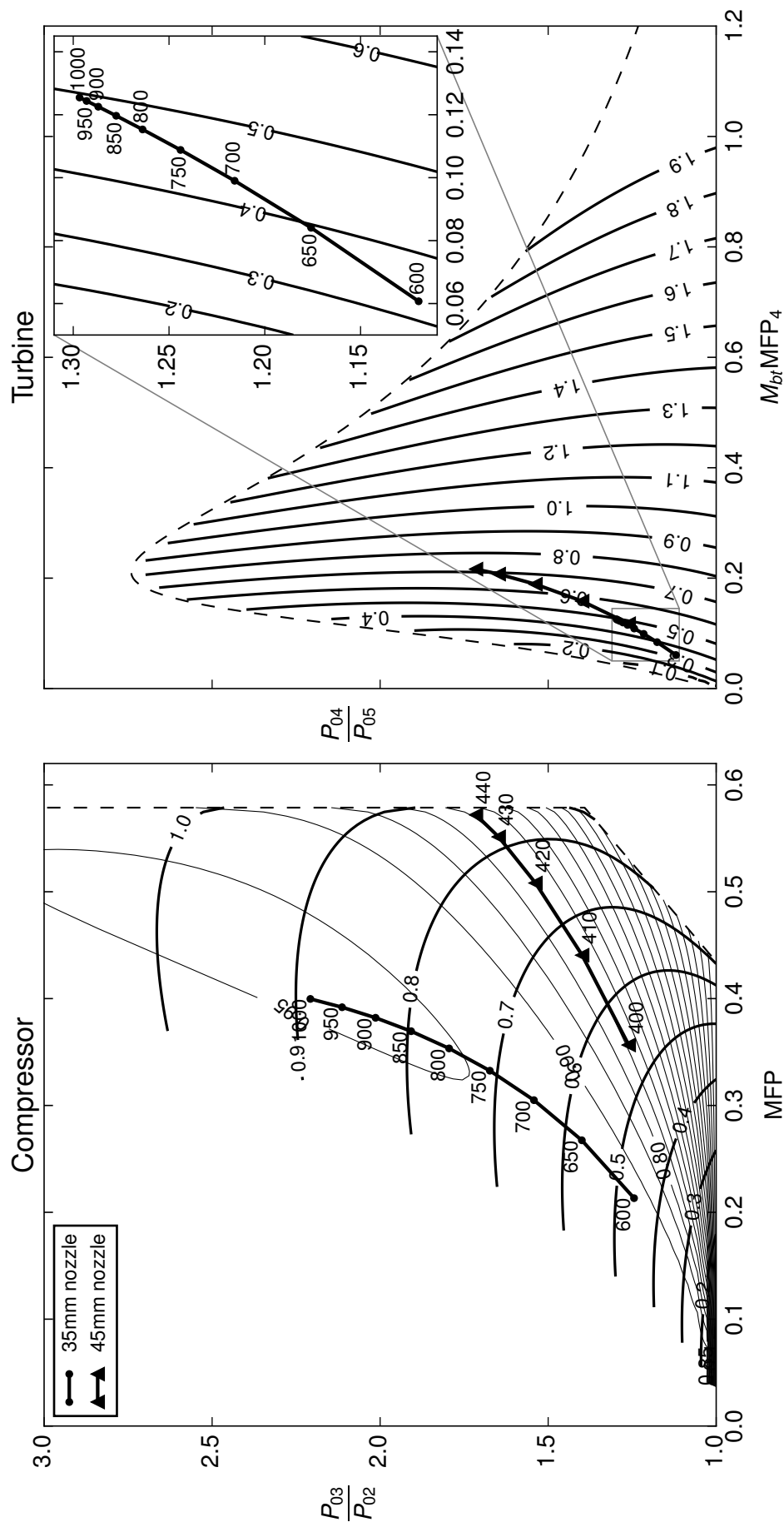
1. the nozzle exit diameter should be reduced to 35mm, and
2. the turbine areas should be redesigned. The inlet area should be made smaller so that it chokes in the design point, and the exit area should be made slightly larger than the inlet, to allow inlet choke and keep velocities approximately constant throughout the turbine.

Off course, the efficiencies are grossly overestimated in this model. Further refinements might change some of these remarks.

Figure 6 shows the temperatures, pressures and Mach numbers for each engine station at a reference turbine inlet temperature of 1000K and using the redesigned 35mm nozzle. Again, it shows that the turbine inlet is very far from choke. This is a serious design flaw. The redesigned nozzle is almost choked in this condition, which is what is desired. Perhaps it could be made even narrower, to increase its Mach number to unity and thus generate more thrust. This, however, would decrease the surge margin.

Figure 7 shows a comparison of the engine model (considering the installed 45mm nozzle) and experimental data obtained in a test bench by Oliveira (2017b). In a qualitative analysis,

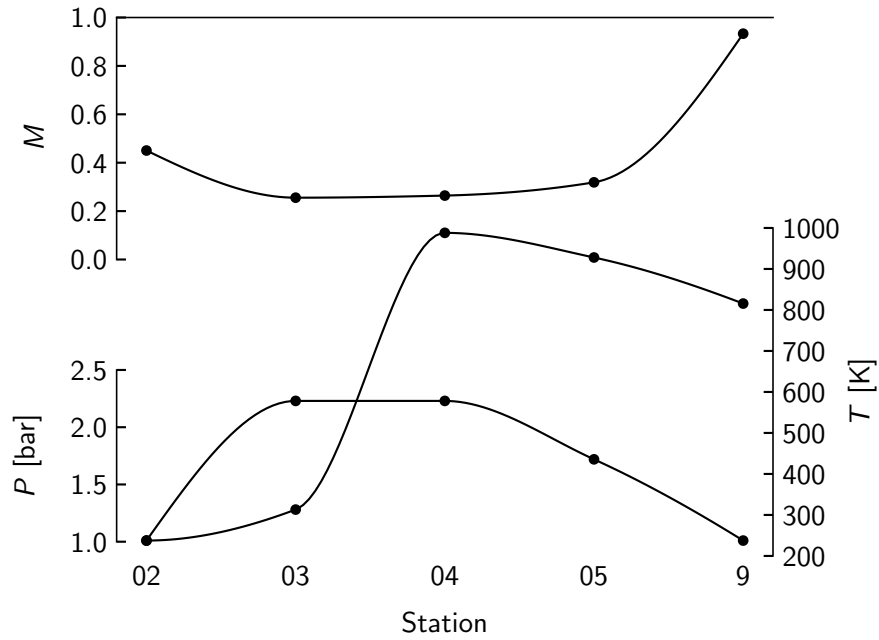
Figure 5 – Component maps and work line of the VT-80 engine



Source: author's figure.

Compressor and turbine maps for the VT-80 engine, with the work lines superimposed (thick line). The medium thickness contours are of constant blade Mach number and the thin contours are of constant polytropic efficiency. Two working lines are shown: one for the actual turbine nozzle with 45mm diameter and another for a redesigned nozzle with 35mm diameter. They are labeled with the turbine inlet stagnation temperatures ( $T_{04}$ ), in kelvin. The surge line is not explicitly shown, and is the limit where the isospeed lines become undefined to the left hand side of the figure.

Figure 6 – Flow state at each engine station

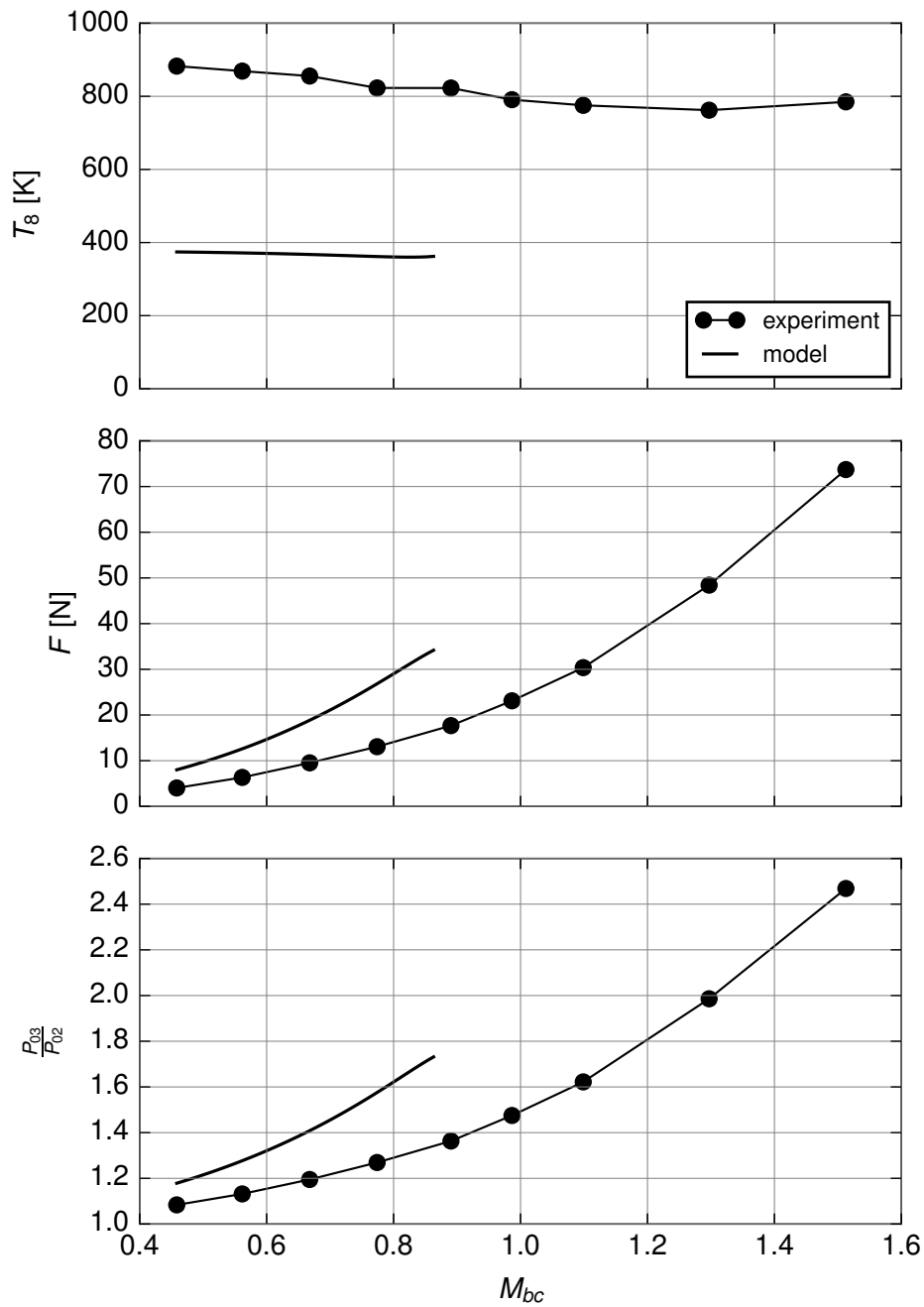


Source: author's figure.

Temperature, pressure and Mach number for each engine station at a turbine inlet temperature of 1000K and flight Mach number zero, using the redesigned 35mm nozzle. The temperature and pressure scales were chosen to keep their ratios to the reference values  $P_{01}$  and  $T_{01}$  comparable. When the station number is prefixed with a "0", the properties are from stagnation, otherwise they are static. Interpolation was done with the PCHIP algorithm (FRITSCH; CARLSON, 1980), to keep lines smooth while not introducing artificial maxima.

the curve shapes are quite similar. Pressure ratio and thrust rise with the square of spool speed and exhaust gas temperatures are flat for both cases. The large quantitative difference in this temperature between model and experimental data is explained by the very low  $T_{04}$  of the model, as can be seen in Figure 5. Even then the model shows greater thrust and pressure ratio. This is due to the very high efficiency considered by the model, since many losses were not included.

Figure 7 – Comparison of model and experimental data



Source: author's figure.

The performance predicted by the semi-empirical model data shows the trends exhibited by the experimental data, i.e. the Exhaust Gas Temperature (EGT) is mostly flat and the thrust ( $F$ ) and compressor pressure ratio ( $\frac{P_{03}}{P_{02}}$ ) rise quadratically with the compressor tip Mach number ( $M_{bc}$ ). The large (approximately 200%) quantitative error is probably due to the overly optimistic efficiency of the model components. The model data is only available up to the engine inlet choke. Experimental data was obtained by Oliveira (2017b).



## 5 Conclusion

*The Road goes ever on and on  
Down from the door where it began.  
Now far ahead the Road has gone,  
And I must follow, if I can,  
Pursuing it with eager feet,  
Until it joins some larger way,  
Where many paths and errands meet.  
And whither then? I cannot say.*

*-Bilbo Baggins*

This work has shown that it is possible to model a gas turbine engine's static and dynamic behavior in a realistic way from simple unidimensional flow considerations and semi-empirical corrections. This model is superior to the thermodynamic cycle model because it represents phenomena such as surge, choke, spool and gas dynamics which are not captured by the former. It is also an efficient way to review a design before incurring in manufacturing costs. It does, however, requires more refined geometry data than the cycle model, but the data required is generally available very early in the design process, since it is mostly composed of the basic engine areas and angles.

The model suggest in this work is most easily expressed as a implicit nonlinear system of equations. It must be solved numerically. Fortunately, there are many high quality computer codes which can deal with most of the problems encountered. There is not, however, a good partial differential algebraic equation solver readily available, and this proved a challenge for modelling surge. In general, the nonlinear systems proposed can exhibit convergence issues in the lack of appropriate initial guesses when using newton-type solvers. A good alternative is to use a path following solver.

For future work, we recommend the implementation of the remaining loss models for the compressor and of all loss models for the engine. This will allow much more precise estimation of the engine performance, and so the comparative analysis of model predictions and experimental data should be performed again. We believe that the results will be much better. Furthermore, the dynamical engine model should be implemented and tested against experimental data. The surge model using the derivative of the compressor and turbine characteristics should also be implemented, perhaps using some discretization of the map to calculate the partial derivatives or by converting the problem into a regular DAE through the use of the chain rule. After all this is done, the model will be ready for use in improving the VT-80 engine design, designing other engines and also designing and testing their control systems.

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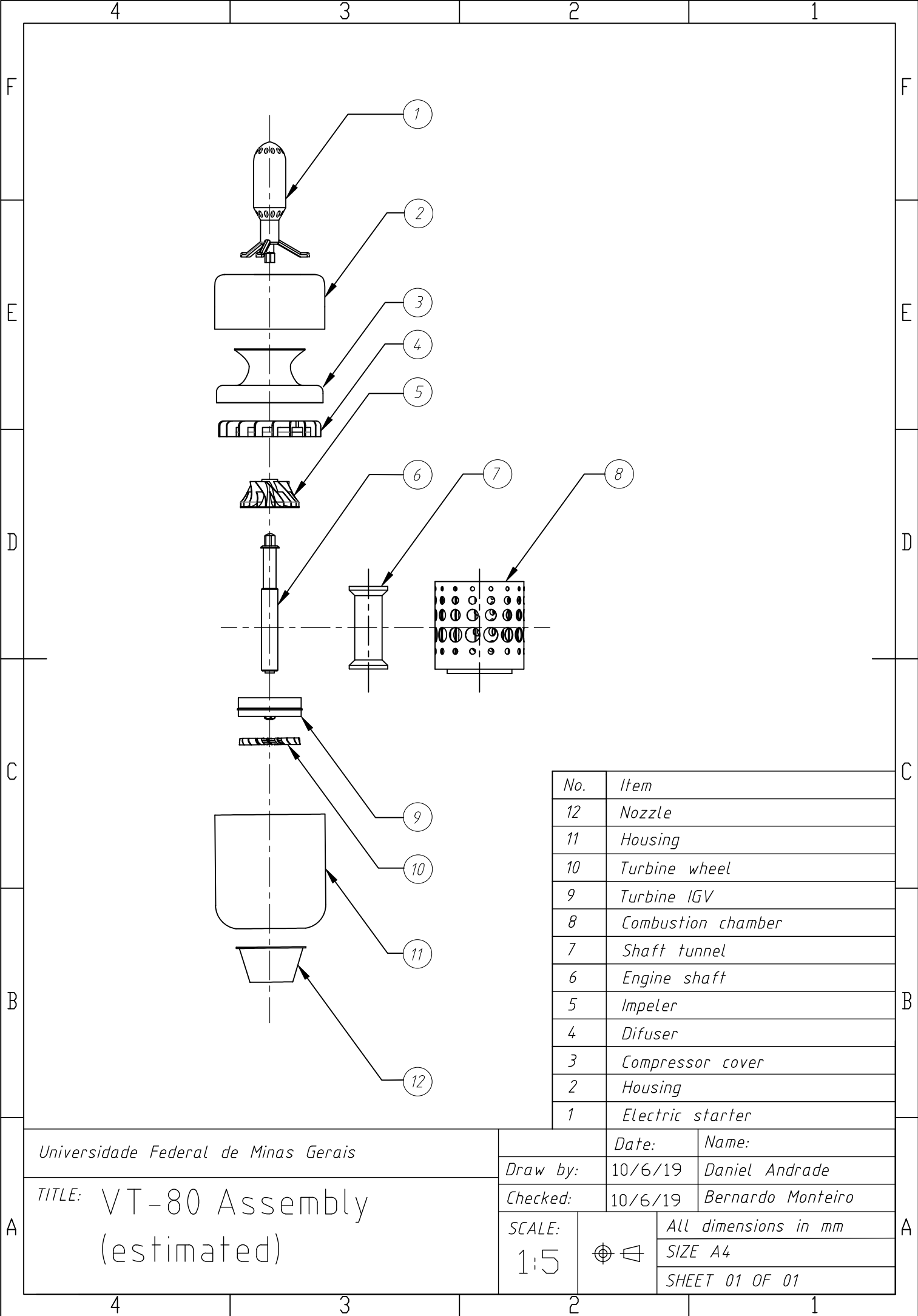
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# Annex

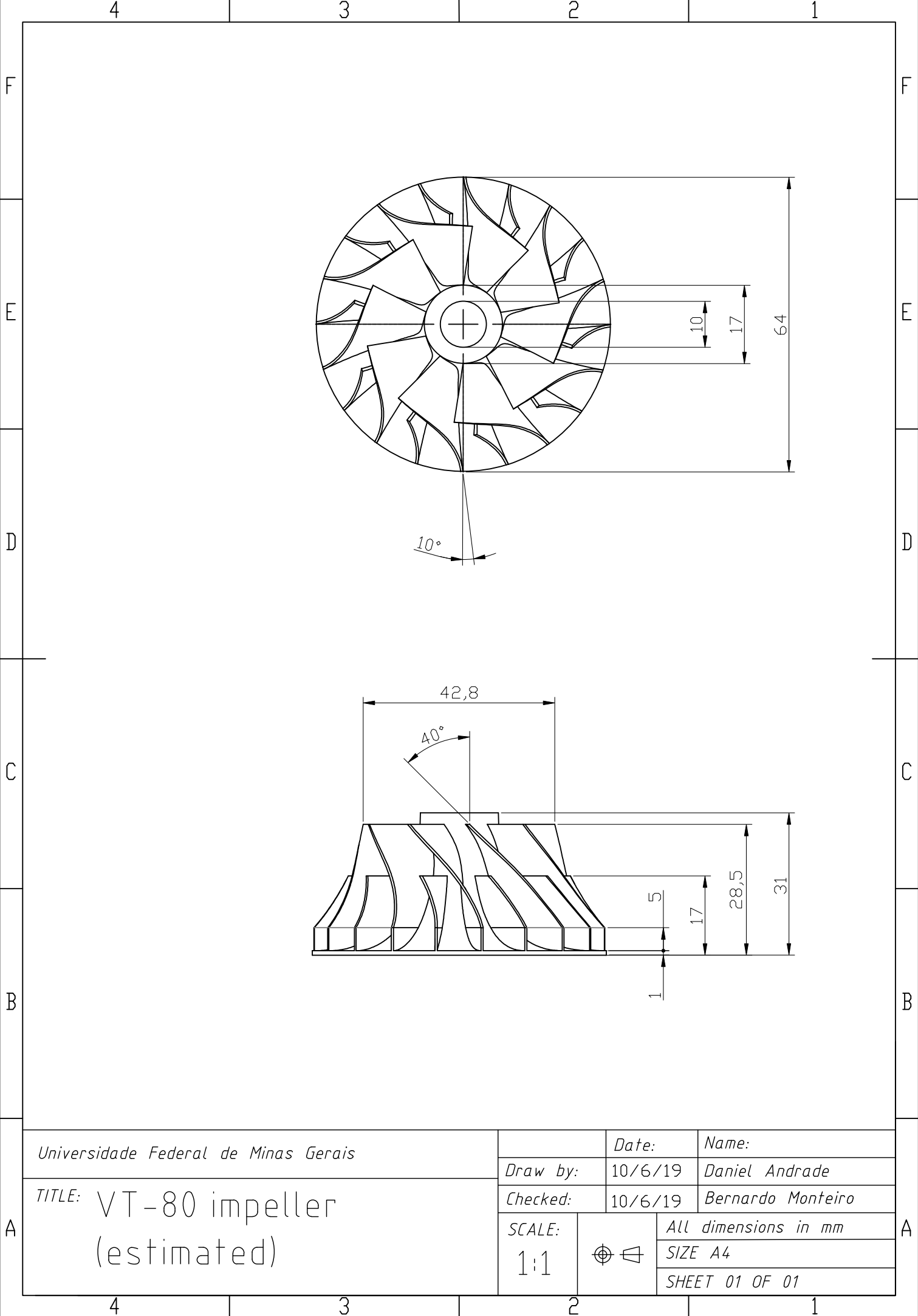


| No. | Item               |
|-----|--------------------|
| 12  | Nozzle             |
| 11  | Housing            |
| 10  | Turbine wheel      |
| 9   | Turbine IGV        |
| 8   | Combustion chamber |
| 7   | Shaft tunnel       |
| 6   | Engine shaft       |
| 5   | Impeler            |
| 4   | Difuser            |
| 3   | Compressor cover   |
| 2   | Housing            |
| 1   | Electric starter   |

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TITLE: VT-80 Assembly  
(estimated)

|          |                           |
|----------|---------------------------|
| Date:    | Name:                     |
| Draw by: | 10/6/19 Daniel Andrade    |
| Checked: | 10/6/19 Bernardo Monteiro |
| SCALE:   | All dimensions in mm      |
| 1:5      | SIZE A4                   |
|          | SHEET 01 OF 01            |



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Date: Name:

Draw by: 10/6/19 Daniel Andrade

Checked: 10/6/19 Bernardo Monteiro

TITLE: VT-80 impeller  
(estimated)

SCALE:

1:1

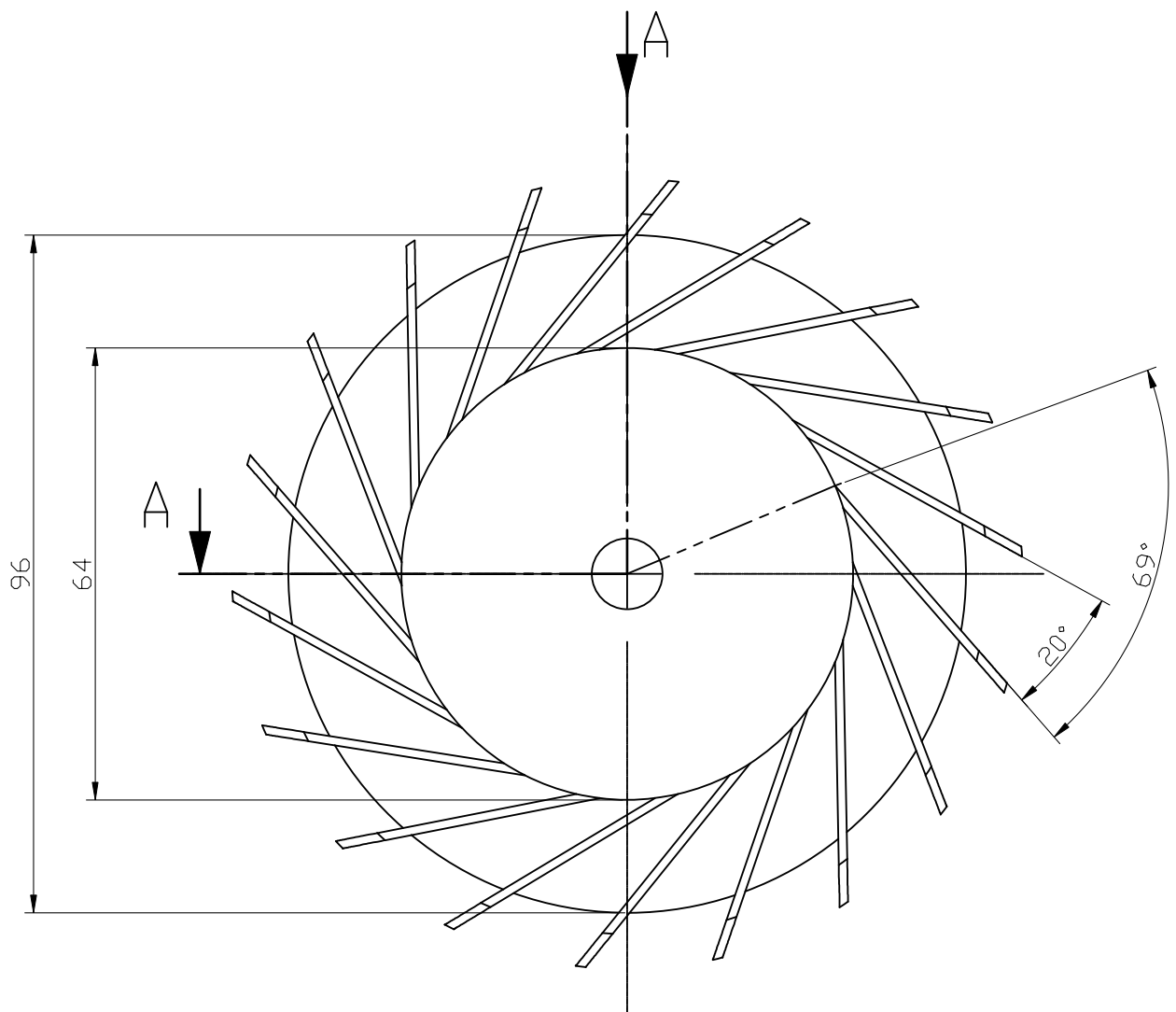


All dimensions in mm

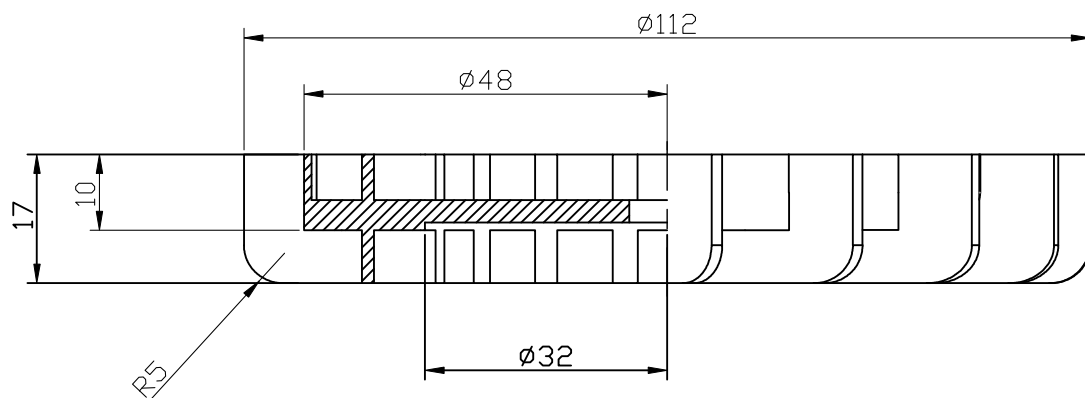
SIZE A4

SHEET 01 OF 01





SECTION A-A  
Scale 1:1



Universidade Federal de Minas Gerais

TITLE: VT-80 Difuser  
(estimated)

Draw by:

Date:

Name:

Checked:

10/6/19

Daniel Andrade

SCALE:

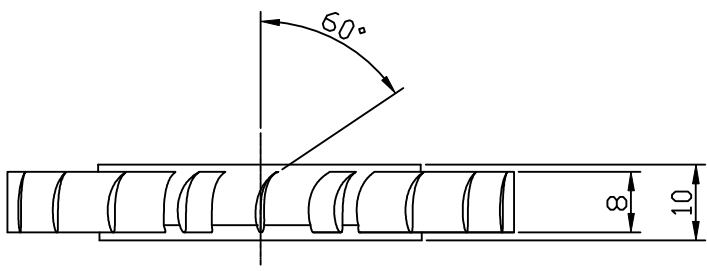
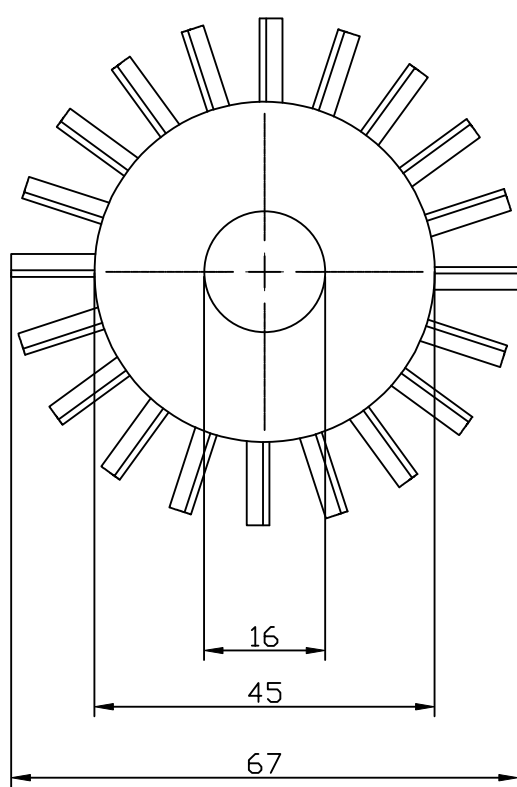
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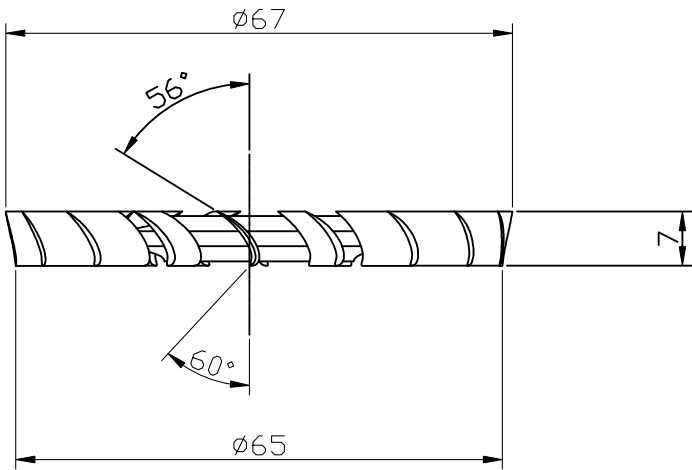
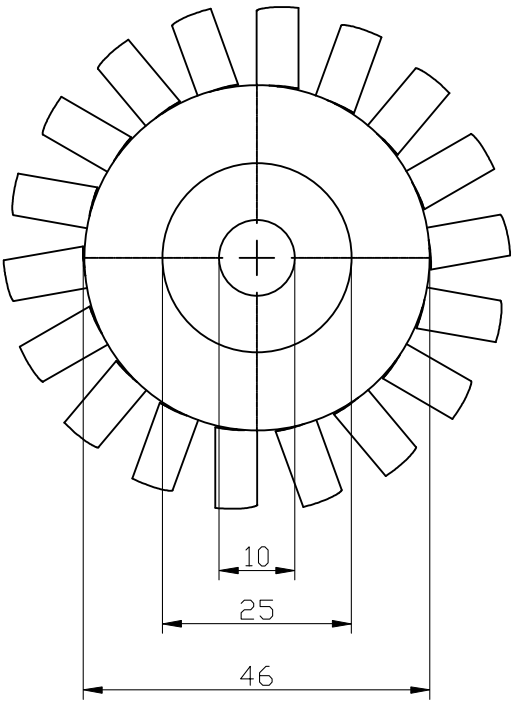
All dimensions in mm

SIZE A4

SHEET 01 OF 01



|                                                     |                                      |  |                      |                   |
|-----------------------------------------------------|--------------------------------------|--|----------------------|-------------------|
| <div>TITLE: VT-80 Turbine IGV<br/>(estimated)</div> | Universidade Federal de Minas Gerais |  | Date:                | Name:             |
|                                                     | Draw by:                             |  | 10/6/19              | Daniel Andrade    |
|                                                     | Checked:                             |  | 10/6/19              | Bernardo Monteiro |
|                                                     | SCALE:                               |  | All dimensions in mm |                   |
| 1:1                                                 |                                      |  | SIZE A4              |                   |
|                                                     |                                      |  | SHEET 01 OF 01       |                   |



Universidade Federal de Minas Gerais

TITLE: VT-80 Turbine wheel  
(estimated)

Draw by:

Date:

Name:

10/6/19

Daniel Andrade

Checked:

10/6/19

Bernardo Monteiro

SCALE:

1:1



All dimensions in mm

SIZE A4

SHEET 01 OF 01